

Leakage-safe similarity-neighbour side information for graph-based drug-target interaction prediction

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Abstract

[TODO: Write only after all fixed-fold results, effect sizes and limitations are finalized.]

1 Introduction

Inferring drug-target interactions (DTIs) from molecular and protein information can prioritize candidate mechanisms for drug discovery and repurposing. Data-driven predictors have therefore progressed from paired sequence or chemical representations, exemplified by DeepDTA, to graph-based drug encoders and attention-based interaction models [1, 2, 3]. These approaches have established useful representation-learning routes, but their reported performance remains strongly coupled to what evidence is allowed to cross an evaluation split.

This issue is particularly consequential for binary DTI prediction. A model may perform well when a test pair contains a drug or a protein that is well represented in the training graph, while its utility is less clear when a drug, protein or pair is held out. Similarity between compounds or proteins can be valuable for these settings: earlier predictors have used neighbourhood regularization and heterogeneous network integration to incorporate related drug and protein evidence [4, 5]. However, label-derived neighbour summaries must be constructed from the training partition alone. Otherwise, a held-out DTI label can enter either message passing or a candidate-pair feature indirectly.

Here, we study a leakage-safe use of similarity-neighbour information in a graph-based DTI predictor. The selected model, ArnoldiGCL-S1/S2, combines separate drug and protein feature projections, a training-fold structural encoder and seven side-information descriptors derived solely from positive training DTIs. Rather than using similarity only as a regularizer or a network integration mechanism, this design exposes training-fold neighbour evidence as candidate-pair descriptors that can be ablated and shuffled. We evaluate it against feature-only, graph-based and bilinear-attention baselines under fixed random, cold-drug, cold-protein and cold-pair splits. The evaluation is designed to identify not only gains from aligned neighbour information, but also settings in which the same information does not improve over simpler alternatives.

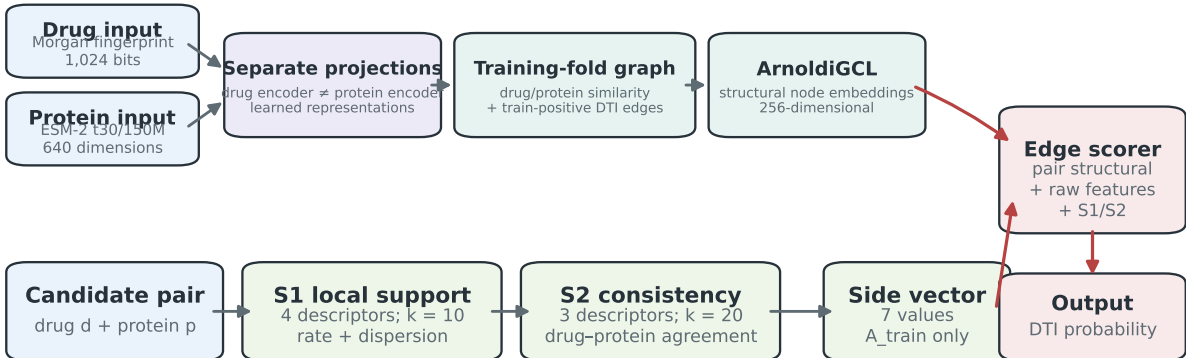
2 Methods

2.1 Problem setting and feature representations

We considered binary drug-target interaction prediction for a drug set $\mathcal{D}$ and a protein set $\mathcal{P}$. Let $A \in \{0, 1\}^{|\mathcal{D}| \times |\mathcal{P}|}$ denote the observed interaction matrix, where $A_{ij} = 1$ indicates an annotated interaction between drug i and protein j . Each drug was represented by a cached 1,024-bit radius-2 Morgan bit vector [6], and each protein by a cached mean-pooled ESM-2 t30/150M embedding (640 dimensions) [7]. The drug and protein feature encoders were distinct learnable projection networks; their parameters were not shared.

Leakage-safe ArnoldiGCL with similarity-neighbour side information

The drug and protein pathways remain separate; only training-fold evidence enters graph and side-information construction.



Leakage control: graph edges and S1/S2 statistics use positive DTI labels from A_{train} only; validation/test labels are excluded.

Encoders are not shared

Figure 1: Overview of the selected ArnoldiGCL-S1/S2 DTI predictor. Drug Morgan fingerprints and protein ESM-2 embeddings enter distinct learnable projection networks; the encoder weights are not shared. A training-fold graph supplies structural embeddings, while the S1/S2 branch derives seven candidate-pair descriptors from feature-based neighbours and positive A_{train} edges only. Structural pair features, raw features and side information are then combined by an edge scorer. Validation and test DTI labels are excluded from graph and side-information construction.

We constructed a drug similarity matrix S_D and a protein similarity matrix S_P by cosine similarity of the corresponding cached feature vectors. These similarities were feature-derived and were not calculated from DTI labels.

2.2 Leakage-safe graph construction

For every outer fold, the graph contained only positive DTI edges from the training partition. Drug-drug and protein-protein similarity edges were also restricted to entities observed in that training partition. The default graph connected the top 10 drug neighbours and top 10 protein neighbours; all graph edges were represented bidirectionally. The threshold-GCN baseline instead retained within-type similarity edges whose similarity exceeded 0.3. No validation or test DTI labels were added to a graph or side-information feature.

2.3 ArnoldiGCL structural representation

The projected drug and protein features were concatenated as nodes in the training-fold graph and passed to the ArnoldiGCL encoder. The structural embedding dimension was 256. The downstream edge scorer received the structural embeddings of a candidate drug-protein pair together with the respective raw Morgan and ESM-2 representations. The encoder was optimized for 50 epochs, and each edge scorer was optimized for 200 epochs with Adam (learning rate 10^{-3} and dropout 0.5). For encoder pretraining, the objective was binary discrimination between the graph representation of the projected node features and a negative view formed by a random permutation of node features. Each update used two positive and two negative node-level logit sets, optimized with binary cross-entropy.

The drug and protein projections remained separate throughout this objective.

2.4 Leakage-safe S1/S2 side information

For a candidate pair (i, j) , all side information was computed from the binary matrix A_{train} that contains positive training DTI edges only. Let $N_D^k(i)$ and $N_P^k(j)$ be the top- k most similar training drugs and proteins, respectively. A queried entity was excluded from its own neighbourhood.

The S1 module used $k = 10$ and produced four descriptors: the mean and standard deviation of $A_{d,j}$ over $d \in N_D^{10}(i)$, and the mean and standard deviation of $A_{d,p}$ over $d \in N_D^{10}(i), p \in N_P^{10}(j)$. Thus, S1 summarizes local neighbour support and its dispersion without accessing the label of the candidate pair.

The S2 module used $k = 20$ and produced three consistency descriptors: the mean of $A_{d,j}$ over $d \in N_D^{20}(i)$, the mean of $A_{i,p}$ over $p \in N_P^{20}(j)$, and the mean of $A_{d,p}$ over the corresponding drug and protein neighbour cross-product. The seven S1/S2 values were passed through the side-information branch of the edge scorer. The S1-only and S2-only ablations zero-padded the complementary descriptors; the shuffled control permuted the seven-dimensional side-information vectors within each partition while preserving their marginal distribution.

2.5 Comparators and implementation details

The benchmark includes logistic regression and an MLP on concatenated Morgan and ESM-2 representations, a two-layer GCN on the leak-safe k NN graph, a threshold-similarity GCN comparator, and DrugBAN [1, 2, 3]. All compared methods use the same frozen positive partitions and partition-specific negative examples. This aligns the evaluated drug-protein pairs across methods but does not force identical input modalities: DrugBAN consumes canonical SMILES-derived graphs and protein sequences, whereas the feature and graph comparators use the frozen Morgan and ESM-2 representations described above. DrugBAN is consequently reported as a matched-partition comparator in this benchmark, not as a claim to reproduce its published benchmark scores. For DrugBAN, deterministic SMILES-to-graph and protein integer preprocessing was cached within a fold when an input recurred; this does not alter the DrugBAN architecture, optimizer or emitted batch values. DrugBAN was trained with Adam at a learning rate of 5×10^{-5} , batch size 64 and a fixed 100-epoch schedule; the validation-AUC checkpoint was evaluated once on the corresponding test partition. All DrugBAN folds used TensorFloat-32 matrix operations on RTX 4080 GPUs; this execution setting was recorded in every result artifact and held fixed across B5 folds. The two GCN baselines use cached sparse normalized aggregation on BindingDB to bound peak memory. This implementation was verified against the original dense aggregation for forward output and backward gradients, and was smoke-tested on the full BindingDB threshold graph before running the queue.

3 Experimental protocol

3.1 Datasets and frozen evaluation splits

We evaluate BioSNAP, Human and BindingDB under four scenarios: random, cold-drug, cold-protein and cold-pair splits. Each scenario uses three fixed random seeds and five outer folds, yielding 15 held-out evaluations per dataset-scenario-method combination. Within each outer training partition, approximately 10% of eligible training positives are reserved for model selection. For cold-drug and cold-protein splits, validation and test entities are disjoint from training entities. For

Family	Model	Information available to the model
Feature-only	Logistic regression	Concatenated Morgan and ESM-2 pair features.
Feature-only	MLP	Concatenated Morgan and ESM-2 pair features.
Graph-based	GCN	Fold-local positive DTI graph with top-10 within-type similarity neighbours.
Graph-based	Threshold GCN	Fold-local positive DTI graph with within-type similarity edges above 0.3.
Bilinear-attention	DrugBAN	Canonical molecular graph and protein-sequence inputs, trained on the same frozen pair partitions.
Ablation	ArnoldiGCL w/o S1/S2; +S1; +S2; shuffled S1/S2	The same fold-local ArnoldiGCL structural encoder, with the indicated side-information treatment.
Full model	ArnoldiGCL-S1/S2	Structural pair features, raw Morgan and ESM-2 features, and aligned seven-dimensional S1/S2 information.

Table 1: Prespecified comparator families. All learning, graph construction and label-derived side information are restricted to the outer training fold.

cold-pair splits, train, validation and test positives are sampled from disjoint diagonal drug–protein blocks; off-block pairs are not evaluated.

The three local `full.csv` files were verified as exact Git-blob matches to the pinned DrugBAN dataset snapshot (commit 9923f8c99959e00263103ff9ac61ba0eaccc8e02). Its dataset documentation identifies BindingDB, BioSNAP as distributed by MolTrans, and Human as distributed by TransformerCPI [3, 8, 9, 10]. The generated dataset manifest records this source snapshot, byte-level checksums and the observed input schema for every run.

For every partition, up to 10 unique unannotated pairs were sampled per positive without replacement. Test and validation negatives were allocated before training negatives to make held-out samples invariant to later training-set changes. In unusually dense blocks with fewer available unique negatives, all available unique negatives were used and the realized partition size was retained in the fold-level result record.

3.2 Comparator families

Table 1 distinguishes feature-only, graph-based, bilinear-attention and ablation controls. All rows use exactly the same frozen positive partitions and partition-specific negative examples. “Threshold GCN” is a comparator implemented with the specified 0.3 within-type similarity threshold; it is not presented as a full reproduction of an external method.

3.3 Evaluation and statistical reporting

We report area under the receiver operating characteristic curve (AUC), area under the precision–recall curve (AUPR), F1 score and accuracy. Checkpoints were selected by validation AUC; F1 thresholds were selected by maximal validation F1 and applied once to the test partition. The primary tables will report mean $\pm$ sample standard deviation across 15 fixed seed–fold evaluations. Pairwise comparisons will be paired by identical seed–fold identity and summarized with effect sizes and two-sided Wilcoxon signed-rank statistics. Because folds sharing a random seed are not independent biological or experimental replicates, these p -values are descriptive summaries of paired fold differences rather than confirmatory population-level inference. For the complete family of reported descriptive comparisons, two-sided p -values will be adjusted using the Benjamini–Hochberg procedure; both raw and adjusted values will be provided as source data.

3.4 Ablation and robustness analyses

We compare the full S1/S2 model with a structural-only version, S1-only and S2-only variants, and a partition-wise shuffled S1/S2 control. The shuffled control distinguishes the presence of extra numerical inputs from alignment between side information and the evaluated drug-protein pair.

[TODO: Insert final dataset statistics from the completed artifacts.]

3.5 Compute environment and reproducibility

The principal benchmark was executed on NVIDIA GeForce RTX 4080 GPUs. Every result artifact records the dataset, scenario, seed, outer fold, split signature, validation criterion, test metrics and realized partition sizes. DrugBAN artifacts additionally record the fixed 100-epoch schedule, input preprocessing cache and TensorFloat-32 setting. [TODO: Insert the final wall-clock summary after the queue completes.]

4 Results

4.1 Complete benchmark comparison

[TODO: Results are generated only after the verified aggregate CSV is complete. Every central claim must state dataset, split, metric, comparator, mean difference and $n = 15$ folds.]

4.2 Contribution of S1 and S2

[TODO: Populate from the complete ablation figure and paired analysis. Do not interpret a shuffled-control difference as causal or biological evidence.]

4.3 Where the method does not improve

[TODO: Report scenario-specific ties or losses against MLP, GCN, threshold-GCN or DrugBAN in the main text. This section is mandatory for claim calibration.]

5 Discussion

[TODO: Interpret the complete benchmark, including scenarios with no gain over simpler baselines.]

6 Data and code availability

Generated benchmark artifacts. The frozen train, validation and test partitions, per-fold result JSON files, aggregate tables, figure source data and the dataset manifest generated in this study will be deposited in [TODO: repository name and persistent identifier] before publication. The manifest records the SHA256 checksum, Git-blob checksum, observed schema and pinned upstream snapshot of every input CSV used for the benchmark.

Reused input data. The benchmark uses local CSV files labelled BioSNAP, Human and BindingDB. Each file was verified byte-for-byte against the corresponding `full.csv` file in the DrugBAN repository dataset snapshot (commit 9923f8c99959e00263103ff9ac61ba0eaccc8e02); that snapshot documents BindingDB, MolTrans/BioSNAP and TransformerCPI/Human as the respective sources [3, 8, 9, 10]. The DrugBAN repository is MIT-licensed; original data-provider terms govern reuse and redistribution, so this manuscript does not claim that the local copies may be redistributed.

Code availability. The frozen split generator, benchmark runners, plotting scripts, manuscript generators and final audit script will be made available at [TODO: code repository and commit or release tag]. The release will include environment specifications and commands required to regenerate the reported tables and figures from the archived result artifacts.

Declarations

Author contributions. [TODO: To be completed by the authors.]

Competing interests. [TODO: To be completed by the authors.]

References

- [1] Hakime Öztürk, Arzucan Özgür, and Elif Ozkirimli. DeepDTA: deep drug–target binding affinity prediction. *Bioinformatics*, 34(17):i821–i829, 2018.
- [2] Thin Nguyen, Hang Le, Thomas P. Quinn, Tri Nguyen, Thuc Duy Le, and Svetha Venkatesh. GraphDTA: predicting drug–target binding affinity with graph neural networks. *Bioinformatics*, 37(8):1140–1147, 2021.
- [3] Peizhen Bai, Filip Miljković, Bino John, and Haiping Lu. Interpretable bilinear attention network with domain adaptation improves drug–target prediction. *Nature Machine Intelligence*, 5(2):126–136, 2023.
- [4] Yong Liu, Min Wu, Chunyan Miao, Peilin Zhao, and Xiao-Li Li. Neighborhood regularized logistic matrix factorization for drug–target interaction prediction. *PLOS Computational Biology*, 12(2):e1004760, 2016.
- [5] Yunan Luo, Xinbin Zhao, Jingtian Zhou, Jinglin Yang, Yanqing Zhang, Wenhua Kuang, Jian Peng, Ligong Chen, and Jianyang Zeng. A network integration approach for drug–target interaction prediction and computational drug repositioning from heterogeneous information. *Nature Communications*, 8(1):573, 2017.
- [6] David Rogers and Mathew Hahn. Extended-connectivity fingerprints. *Journal of Chemical Information and Modeling*, 50(5):742–754, 2010.
- [7] Zeming Lin, Halil Akin, Roshan Rao, Brian Hie, Zhongkai Zhu, Wenting Lu, Nikita Smetanin, Robert Verkuil, Ori Kabeli, Yaniv Shmueli, Allan dos Santos Costa, Maryam Fazel-Zarandi, Tom Sercu, Salvatore Candido, and Alexander Rives. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637):1123–1130, 2023.
- [8] T. Liu, Y. Lin, X. Wen, R. N. Jorissen, and M. K. Gilson. BindingDB: a web-accessible database of experimentally determined protein-ligand binding affinities. *Nucleic Acids Research*, 35(Database):D198–D201, 2007.
- [9] Kexin Huang, Cao Xiao, Lucas M. Glass, and Jimeng Sun. MolTrans: Molecular interaction transformer for drug–target interaction prediction. *Bioinformatics*, 37(6):830–836, 2021.
- [10] Lifan Chen, Xiaoqin Tan, Dingyan Wang, Feisheng Zhong, Xiaohong Liu, Tianbiao Yang, Xiaomin Luo, Kaixian Chen, Hualiang Jiang, and Mingyue Zheng. TransformerCPI: improving compound–protein interaction prediction by sequence-based deep learning with self-attention mechanism and label reversal experiments. *Bioinformatics*, 36(16):4406–4414, 2020.